

HTML5 Exercises

Project Theme: Local Community Event Portal

Exercise 1: HTML5 Base Template

Set up the base document structure that every page on the portal will use.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Local Community Event Portal">
  <title>Community Event Portal</title>
</head>
<body>
  <!-- Navigation -->
  <header>
    <nav>
      <a href="#home">Home</a> |
      <a href="#events">Events</a> |
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <!-- Main Content -->
  <main>
    <section id="home"><h1>Welcome to the Community Portal</h1></section>
    <section id="events"><h2>Upcoming Events</h2></section>
    <section id="contact"><h2>Contact Us</h2></section>
  </main>

  <!-- Footer -->
  <footer>
    <p>&copy; 2025 Local City Council. All rights reserved.</p>
  </footer>
</body>
</html>
```

Exercise 2: Navigation and Linking

Provide intuitive navigation between Home, Events, and Contact sections.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Community Portal - Navigation</title>
</head>
<body>
  <header>
    <nav>
      <a href="#home">Home</a> |
      <a href="#events">Events</a> |
      <a href="#contact">Contact</a> |
      <a href="help.html" target="_blank">Help</a>
```

```

        </nav>
    </header>
    <main>
        <section id="home">
            <h1>Home</h1>
            <p>Welcome to the Local Community Event Portal.</p>
        </section>
        <section id="events">
            <h2>Events</h2>
            <p>Browse upcoming community events here.</p>
        </section>
        <section id="contact">
            <h2>Contact</h2>
            <p>Email: council@city.gov | Phone: (555) 000-1234</p>
        </section>
    </main>
</body>
</html>

```

Exercise 3: Welcome Message with Styling and ID/Class

Display a welcome banner styled uniquely for a logged-in user using id and class.

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Welcome Banner</title>
    <style>
        #welcomeBanner {
            background-color: blue;
            color: white;
            padding: 20px;
            font-size: 20px;
            border-radius: 8px;
        }
        .highlight {
            font-weight: bold;
            text-decoration: underline;
        }
    </style>
</head>
<body>
    <div id="welcomeBanner">
        Welcome back, <span class="highlight">Alice Johnson</span>!
        <span style="color: red; font-weight: bold;">
            Special Offer: Free entry to next event!
        </span>
    </div>
    <p class="highlight">Highlighted Paragraph</p>
    <p>Regular paragraph without highlight.</p>
</body>
</html>

```

Exercise 4: Image Gallery for Community Events

Show images from past events in a table layout with captions.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Event Gallery</title>
  <style>
    .gallery-img {
      border: 3px solid #333;
      width: 200px;
      height: 150px;
      object-fit: cover;
    }
    table { border-collapse: collapse; margin: auto; }
    td { padding: 10px; text-align: center; }
    caption { font-size: 1.4em; font-weight: bold; margin-bottom: 10px; }
  </style>
</head>
<body>
  <table>
    <caption>Community Events Photo Gallery</caption>
    <tr>
      <td>
        
        <p>Tech Meetup</p>
      </td>
      <td>
        
        <p>AI Conference</p>
      </td>
      <td>
        
        <p>Frontend Bootcamp</p>
      </td>
    </tr>
    <tr>
      <td>
        
        <p>Community Gathering</p>
      </td>
      <td>
        
        <p>Skills Workshop</p>
      </td>
      <td>
        
        <p>Council Open Day</p>
      </td>
    </tr>
  </table>
</body>
</html>

```

Exercise 5: Event Registration Form

Allow residents to register for events using an HTML form with validation.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Event Registration</title>
  <style>
    body { font-family: Arial, sans-serif; max-width: 500px; margin: 40px auto; }
    label { display: block; margin-top: 12px; font-weight: bold; }
    input, select, textarea {
      width: 100%; padding: 8px; margin-top: 4px;
      border: 1px solid #ccc; border-radius: 4px; box-sizing: border-box;
    }
    button {
      margin-top: 16px; padding: 10px 20px;
      background: #2e7d32; color: white;
      border: none; border-radius: 4px; cursor: pointer;
    }
    output { display: block; margin-top: 12px; color: green; font-weight: bold; }
  </style>
</head>
<body>
  <h2>Register for an Event</h2>
  <form id="regForm">
    <label for="name">Full Name</label>
    <input type="text" id="name" name="name" placeholder="Enter your full name"
      required autofocus>

    <label for="email">Email Address</label>
    <input type="email" id="email" name="email"
      placeholder="your@email.com" required>

    <label for="date">Preferred Date</label>
    <input type="date" id="date" name="date" required>

    <label for="eventType">Event Type</label>
    <select id="eventType" name="eventType" required>
      <option value="">-- Select Event --</option>
      <option value="tech">Tech Meetup</option>
      <option value="ai">AI Conference</option>
      <option value="frontend">Frontend Bootcamp</option>
    </select>

    <label for="message">Additional Message</label>
    <textarea id="message" name="message" rows="4"
      placeholder="Any special requirements..."></textarea>

    <button type="submit">Register</button>
    <output id="confirmMsg"></output>
  </form>
  <script>
    document.getElementById('regForm').addEventListener('submit', function(e) {
      e.preventDefault();
      const name = document.getElementById('name').value;
      const event = document.getElementById('eventType').value;
      document.getElementById('confirmMsg').textContent =
```

```

        'Thank you, ' + name + '! You have registered for ' + event + '.'';
    });
</script>
</body>
</html>

```

Exercise 6: Event Feedback with Event Handling

Collect real-time feedback using *blur*, *change*, *click*, *dblclick*, and *key* events.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Event Feedback</title>
  <style>
    body { font-family: Arial, sans-serif; max-width: 520px; margin: 30px auto; }
    input, select, textarea { width: 100%; padding: 8px; margin-top: 4px; box-sizing: border-box; }
    #eventImg { width: 200px; cursor: pointer; transition: transform 0.3s; }
    #eventImg.enlarged { transform: scale(2); transform-origin: top left; }
    #fee, #phoneError, #charCount, #confirm { color: green; font-weight: bold; margin-top: 6px; }
    #phoneError { color: red; }
  </style>
</head>
<body>
  <h2>Event Feedback Form</h2>

  <!-- onblur phone validation -->
  <label>Phone Number:
    <input type="text" id="phone" placeholder="10-digit number"
      onblur="validatePhone()">
  </label>
  <div id="phoneError"></div>

  <!-- onchange dropdown -->
  <label>Select Event:
    <select id="eventSel" onchange="showFee()">
      <option value="">--Select--</option>
      <option value="500">Tech Meetup</option>
      <option value="0">AI Conference (Free)</option>
      <option value="1200">Frontend Bootcamp</option>
    </select>
  </label>
  <div id="fee"></div>

  <!-- onclick image enlarge -->
  

  <!-- key event character counter -->
  <label>Feedback:
    <textarea id="feedback" rows="4" placeholder="Your comments..."
      onkeyup="countChars()"></textarea>
  </label>
  <div id="charCount">0 characters</div>

  <!-- onclick confirm -->
  <button onclick="submitFeedback()">Submit Feedback</button>
  <div id="confirm"></div>

```

```

<script>
  function validatePhone() {
    const p = document.getElementById('phone').value;
    document.getElementById('phoneError').textContent =
      /^\d{10}$/.test(p) ? '' : 'Please enter a valid 10-digit phone number.';
  }
  function showFee() {
    const v = document.getElementById('eventSel').value;
    document.getElementById('fee').textContent =
      v ? 'Registration Fee: Rs. ' + v : '';
  }
  function enlargeImg() {
    document.getElementById('eventImg').classList.toggle('enlarged');
  }
  function countChars() {
    const len = document.getElementById('feedback').value.length;
    document.getElementById('charCount').textContent = len + ' characters';
  }
  function submitFeedback() {
    document.getElementById('confirm').textContent =
      'Thank you for your feedback!';
  }
}
</script>
</body>
</html>

```

Exercise 7: Video Invite with Media Events

Show a short event promo video with media events and unload warning.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Event Promo Video</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; padding: 30px; }
    video { border: 3px solid #2e7d32; border-radius: 8px; }
    #videoStatus { color: green; font-weight: bold; margin-top: 10px; }
  </style>
</head>
<body>
  <h2>Community Event Promo</h2>
  <video id="promoVideo" width="560" height="315"
    controls
    oncanplay="videoReady()"
    onended="videoEnded()">
    <source src="promo.mp4" type="video/mp4">
    <source src="promo.ogv" type="video/ogg">
    Your browser does not support the video tag.
  </video>
  <div id="videoStatus">Loading video...</div>

  <script>
    function videoReady() {
      document.getElementById('videoStatus').textContent =
        'Video ready to play!';
    }
    function videoEnded() {

```

```

        document.getElementById('videoStatus').textContent =
            'Thanks for watching! Register now.';
    }

    let formDirty = false;
    document.addEventListener('input', () => { formDirty = true; });

    window.addEventListener('beforeunload', function(e) {
        if (formDirty) {
            e.preventDefault();
            e.returnValue = 'You have unsaved feedback. Are you sure you want to leave?';
        }
    });
</script>
</body>
</html>

```

Exercise 8: Saving User Preferences

Store and retrieve preferred event type using localStorage and sessionStorage.

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>User Preferences</title>
    <style>
        body { font-family: Arial, sans-serif; max-width: 400px; margin: 40px auto; }
        select, button { padding: 8px 14px; margin-top: 10px; display: block; width: 100%; }
        button { background: #1565c0; color: white; border: none;
            border-radius: 4px; cursor: pointer; }
        #status { color: green; margin-top: 10px; font-weight: bold; }
    </style>
</head>
<body>
    <h2>Event Preferences</h2>
    <label for="pref">Preferred Event Type:</label>
    <select id="pref">
        <option value="">-- Select --</option>
        <option value="tech">Tech Meetup</option>
        <option value="ai">AI Conference</option>
        <option value="frontend">Frontend Bootcamp</option>
    </select>
    <button onclick="savePreference()">Save Preference</button>
    <button onclick="clearPreferences()"
        style="background:#c62828;">Clear Preferences</button>
    <div id="status"></div>

    <script>
        // On page load, retrieve saved preference
        window.addEventListener('load', function() {
            const saved = localStorage.getItem('preferredEvent');
            if (saved) {
                document.getElementById('pref').value = saved;
                document.getElementById('status').textContent =
                    'Loaded saved preference: ' + saved;
            }
            // Store session info
            sessionStorage.setItem('sessionStart', new Date().toISOString());

```

```

});

function savePreference() {
  const val = document.getElementById('pref').value;
  if (val) {
    localStorage.setItem('preferredEvent', val);
    document.getElementById('status').textContent =
      'Preference saved: ' + val;
  }
}

function clearPreferences() {
  localStorage.clear();
  sessionStorage.clear();
  document.getElementById('pref').value = '';
  document.getElementById('status').textContent =
    'All preferences cleared.';
}
</script>
</body>
</html>

```

Exercise 9: Geolocation for Event Mapping

Locate the nearest event to the user using the Geolocation API.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Find Nearby Events</title>
  <style>
    body { font-family: Arial, sans-serif; max-width: 500px; margin: 40px auto; }
    button { padding: 10px 20px; background: #1565c0; color: white;
      border: none; border-radius: 4px; cursor: pointer; font-size: 1em; }
    #result { margin-top: 16px; font-size: 1em; color: #333; }
    .error { color: red; }
    .success { color: green; }
  </style>
</head>
<body>
  <h2>Find Nearby Events</h2>
  <button onclick="findNearby()">Find Nearby Events</button>
  <div id="result">Click the button to detect your location.</div>

  <script>
    const events = [
      { name: "Tech Meetup", lat: 40.7128, lon: -74.0060 },
      { name: "AI Conference", lat: 41.8781, lon: -87.6298 },
      { name: "Frontend Bootcamp", lat: 34.0522, lon: -118.2437 }
    ];

    function getDistance(lat1, lon1, lat2, lon2) {
      const R = 6371;
      const dLat = (lat2 - lat1) * Math.PI / 180;
      const dLon = (lon2 - lon1) * Math.PI / 180;
      const a = Math.sin(dLat/2)**2 +
        Math.cos(lat1*Math.PI/180) * Math.cos(lat2*Math.PI/180) *
        Math.sin(dLon/2)**2;

```



```

    return R * 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1-a));
  }

function findNearby() {
  const el = document.getElementById('result');
  if (!navigator.geolocation) {
    el.innerHTML = '<span class="error">Geolocation not supported.</span>';
    return;
  }
  el.textContent = 'Detecting your location...';
  navigator.geolocation.getCurrentPosition(
    function(pos) {
      const { latitude: lat, longitude: lon } = pos.coords;
      let nearest = events.reduce((prev, curr) =>
        getDistance(lat, lon, curr.lat, curr.lon) <
        getDistance(lat, lon, prev.lat, prev.lon) ? curr : prev
      );
      el.innerHTML =
        '<span class="success">Your Location: ' + lat.toFixed(4) +
        ', ' + lon.toFixed(4) + '<br>' +
        'Nearest Event: ' + nearest.name + '</span>';
    },
    function(err) {
      const msgs = {
        1: 'Permission denied. Please allow location access.',
        2: 'Position unavailable.',
        3: 'Request timed out. Please try again.'
      };
      el.innerHTML = '<span class="error">' +
        (msgs[err.code] || 'Unknown error.') + '</span>';
    },
    { enableHighAccuracy: true, timeout: 10000, maximumAge: 0 }
  );
}
</script>
</body>
</html>

```

Exercise 10: Debugging with Chrome DevTools

Use Chrome DevTools and VS Code features to debug layout issues and script errors.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>DevTools Debugging Demo</title>
  <style>
    /* Intentional style for live DevTools inspection */
    body { font-family: Arial, sans-serif; max-width: 600px; margin: 30px auto; }
    #debugBox {
      background: #e3f2fd;
      padding: 20px;
      border: 2px solid #1565c0;
      border-radius: 6px;
    }
    .broken-style { color: red; } /* Typo – find via DevTools */
  </style>
</head>

```

```
<body>
  <h2>DevTools Debugging Demo</h2>
  <div id="debugBox">
    <p>Open DevTools (F12) and inspect this box.</p>
    <p>Try modifying styles in the Elements panel live.</p>
  </div>

  <button onclick="runDebugScript()">Run Script (check Console)</button>
  <div id="output"></div>

  <script>
    // Console logging for inspection
    console.log('Page loaded at:', new Date().toLocaleTimeString());

    let counter = 0;

    function runDebugScript() {
      counter++;
      // Set breakpoint on next line in DevTools Sources tab
      const msg = 'Button clicked ' + counter + ' time(s)';
      console.log(msg);
      document.getElementById('output').textContent = msg;

      // Simulate variable inspection
      const data = { user: 'Alice', event: 'Tech Meetup', count: counter };
      console.table(data);
    }

    // Error example – catch in Console
    window.addEventListener('error', function(e) {
      console.error('Caught error:', e.message, 'at line', e.lineno);
    });
  </script>
</body>
</html>
```